Computer Science

Summative Assignment

Module code and title	COMP2261 Artificial Intelligence
Academic year	2025-26
Coursework title	Bias in AI coursework
Coursework credits	5 credits
% of module's final mark	25%
Lecturer	Shauna Concannon
Submission date*	2pm Thursday, March 26, 2026
Estimated hours of work	10
Submission method	Ultra (with Turnitin check)

Additional coursework files	*Please list all files provided for the coursework, e.g. datasets, template files* *No associated files.*
Required submission items and formats	*Please list the files that should be submitted, including format* • A video recording of your presentation in MP4 format. • A pdf of your slides. • A Generative AI usage statement (max 150 words) • A jupyter notebook containing your data analysis.

* This is the deadline for all submissions except where an approved extension is in place. For benchtests taking place in practical sessions, the given date is the Monday of the week in which the benchtests will take place.

Late submissions received within 5 working days of the deadline will be capped at 40%. Late submissions received later than 5 days after the deadline will receive a mark of 0. It is your responsibility to check that your submission has uploaded successfully and obtain a submission receipt.

Your work must be done by yourself (or your group, if there is an assigned groupwork component) and comply with the university rules about plagiarism and collusion.

Students suspected of plagiarism, either of published or unpublished sources, including the work of other students, or of collusion will be dealt with according to University guidelines (https://durhamuniversity.sharepoint.com/teams/LTH/SitePages/6.2.4.aspx).

Assignment Introduction

In this coursework, you will create an individual video presentation (8 minutes maximum) that critically examines an AI system design and delivers recommendations based on an ethical assessment.

Submission Guidelines

- A recorded video presentation
  - Duration: **Maximum 8 minutes**
  - You should submit your video presentation via the submission point on Blackboard Ultra.
- PDF of the slides used in the presentation
- A Generative AI usage statement outlining any specific uses of Generative AI in the assignment, including prompts (format: word file / text file / pdf)
  - Word count for the Generative AI usage statement: 150 words max
- A Jupyter notebook with any data exploration/analysis conducted (format: .ipynb)

The presentation should be your own work. If you use any generative AI s (as per the departmental policy), you must clearly reference these. All students should submit a statement summarising if and how they have utilised generative AI for their assignment (up to 150 words maximum).

Content Warning: Sensitive Material
This assignment involves analysing real-world datasets used for content moderation. A selection of datasets will be provided, but you should consider carefully which you want to work with as they may include language that is offensive, harmful, and discriminatory, including profanity and hate speech.

Important Guidance

- Objective analysis: You are expected to treat this data as a technical artifact. You do not need to read the data line-by-line; your focus should be on computational analysis and broad patterns.
- Presentation ethics: When presenting your findings, do not include uncensored slurs or highly graphic content in your slides. Use asterisks (e.g., s***) or refer to "identity-based slurs" generally.
- Wellbeing: The data is based on real world moderation. As such will contain content that has been classified as toxic by annotators / content moderation systems. If you find working on this topic too problematic, please contact the sub-module leader by week 7 to discuss alternatives.

Assignment Overview

Scenario: **Online Community Moderation AI**

You are part of the Trust & Safety team at *CivicSphere*, a company developing a machine learning content moderation tool for an online platform. The company aims to

create an automated system that can detect and filter messages containing harmful content to make online spaces safer and more inclusive.

CivicSphere is a startup dedicated to pro-social moderation. Unlike traditional platforms that simply delete ‘bad’ words, CivicSphere aims to understand context distinguishing between a words used to cause offence and the same word used in a self-empowering way by a community. As a Junior Auditor, your job isn't just to make the system ‘accurate’, it’s to make it just. Your goal is to understand why systems may make ‘Quiet Errors’, mistakes that don't crash the system but do alienate specific groups of users.

Harmful content refers to content which causes harm to people, such as offensive or toxic language, aggressive and demeaning content. You can approach toxicity as one overarching category to include any content that may cause harm including hate speech, or constrain this to a particular subtype, for example, profanity and swearing, insults, sexist or ableist content.

However, concerns about potential ethical implications, fairness and bias considerations have been raised during the project kick-off meeting – from issues pertaining to free speech through to what is considered harmful content and ensuring that the system doesn’t unfairly discriminate against particular groups.

You have been tasked with conducting a conceptual stakeholder analysis informed by Value Sensitive Design, auditing available text classification datasets for toxicity detection, and proposing a development plan that ensures the system is fair, ethical, and effective for diverse users.

Dataset & tasks
The Jigsaw Unintended Bias in Toxicity Classification Kaggle dataset is a widely used toxicity dataset that has been used to train toxicity classifiers. You are to perform both an empirical audit, and a critical assessment of the data documentation. You should create a presentation summarising your focused stakeholder analysis and dataset evaluation, and proposing actionable recommendations informed by your findings.

Data files and sensitive content options
You are free to work directly with data available via the Kaggle link but as the dataset is large a sample of the data is provided, that follows a similar distribution.

The original and sample datasets both contain offensive content that can be distressing to work with. You are not required to work with the text comments directly to complete the assignment. It is possible to complete the assignment using the data file ‘toxicity_sample_no_text.csv’ which does not include the actual comments, but all other relevant metadata. I have included a range of data files so that you can pick an approach that feels best for you.

Materials	Description
sample_no_comment_text.csv	This includes all of the sampled data but without the actual text comments, as is the best option to explore representativeness and toxicity classifications without needing to look at the text itself.
sample_data.csv	This includes all of the sampled data.

sample_redacted.csv	This includes the text but has applied a simple profanity classifier to the text, however, it is likely still to contain offensive content.
data_load_redact.ipynb	Example code to load the data and the option to create your own list of words to redact from the comments.

Key Components

Your video presentation should include the following sections:

Introduction [10 marks]

- Present one key ethical challenge in content moderation.
- Introduce the Value Sensitive Design framework and its relevance to the task.

Stakeholder Analysis [25 marks]

Using the Value Sensitive Design approach:

- Identify and analyse 2-3 stakeholders
- For each stakeholder, describe:
  - Their key values (focus on most important, context specific ones)
  - Potential risks/harms they may experience from misclassification or design choices.
- Present one significant value tension between stakeholders illustrated with a concrete example.
- Propose one strategy to address this tension, acknowledging any trade-offs.

Dataset Audit [35 marks]

Critically evaluate the data requirements for training the system. Use simple quantitative analysis (e.g. counts, distributions, or basic visualisations) and critical analysis of documentation.

- Present a critique of one dataset from the list provided on Blackboard.
- Perform a simple data exploration (e.g. class balance, subgroup distributions where available) with a critical reading of any accompanying documentation.
- Compute and present at least two descriptive statistics or visualisations that help reveal potential sources of bias (e.g. label distribution by identity category, prevalence of toxicity labels in subcategories).
- Compare your empirical findings with claims or omissions in the dataset's documentation.
- Discuss two critical data quality considerations including one specific example of potential bias, supported by evidence from your exploratory analysis of the dataset.
- Produce a dataset audit table to display in your presentation.
  - Note: The table can be one slide with brief bullet fragments, not full paragraphs. You are not expected to exhaustively answer each question, just provide 1–2 key points per row. You can draw on your own analysis and the documentation for this.
  - Example analysis dimensions provided below:

Dimension	**Guidance for Analysis**

Provenance & Context	Where did this text originate? How does the platform's culture influence the 'toxicity' found there?
Labelling Process	Who were the annotators? What specific guidance did they receive and what are the implications of this?
Representational Gaps	Which demographic groups are missing or under-represented?
Empirical Finding	Summarise the primary bias or imbalance you discovered through your quantitative analysis.

Recommendations [20 marks]
Provide two specific recommendations that respond to your analysis:

- At least one recommendation should be directly informed by patterns you observed in the dataset (e.g. particular groups disproportionately flagged as toxic).
    - E.g. based on your analysis of existing data you could describe key guidance or processes to inform data collection / preparation / monitoring for this project.
- Note key limitations of your recommendations

Presentation Quality [10 marks]

- Utilises clear visuals to enhance the presentation
- Well-structured with good organisation of materials; all sources are appropriately referenced.
- Concise, engaging content within the time limit.

Marking Scheme

First Class (70%+)

- Shows deep understanding of ethical issues
- Excellent analysis of carefully chosen stakeholders
- Clear, specific examples throughout
- Insightful use of empirical analysis and the audit table to support ethical arguments
- Strong connections between all sections
- Professional delivery within time limit
- Recommendations clearly flow from analysis

Upper Second (60-69%)

- Good grasp of ethical issues
- Well-chosen stakeholders with good analysis
- Good use of examples
- Good use of empirical analysis and the audit table to support ethical arguments
- Most sections connect well
- Clear delivery, good timing
- Sensible recommendations

Lower Second (50-59%)

- Basic understanding shown

- Covers main stakeholders
- Some examples given
- Some empirical analysis and use of the audit table, but lacks depth or clear links to ethical arguments
- Some connections between sections
- Acceptable delivery
- Basic recommendations

Third (40-49%)

- Limited understanding
- Basic stakeholder coverage
- Few specific examples
- Empirical analysis absent or very basic; little/no use of audit table or evidence for arguments
- Weak connections
- Poor timing/delivery
- Vague recommendations

Fail (Below 40%)

- Poor understanding
- Missing key elements
- No specific examples
- No meaningful empirical analysis or audit table
- Disconnected sections
- Very poor timing/delivery
- Missing/inappropriate recommendations